HW03

Games I (P.1)

(A) $x=1,\cdots 9$; left to right; $y=0,\cdots 9$; bottom to top. Red moves toward increasing y

Piece	Position	Legal destinations/directional count.	Count.
Left rook	(1,0)	(1,1) (1,2) (1,3)	3
Right rook	(9,0)	(9,1), (9,2)	2
Left horse	(2,0)	(1,2), (3,2)	2
Right horse.	(8,0)	(7,2) (9,2)	2
Left elephant	(3,0)	(1,2) (5,2)	2
~~Right~~ Left elephant	(7,0)	(5,2) (9,2)	1
Left advisor	(4,0)	(5,1)	1
Right advisor	(6,0)	(5,1)	1
General.	(5,0)	(5,1)	13
Left cannon	(4,2)	Left 3, right 3, down 1, up 6	
Right cannon	(8,2)	Left 3, right 1, down 1, up 4 noncaptures +1 Capture.	10
Soldier 1	(1,4)	(1,5)	1
Soldier 2	(3,3)	(3,4)	1
Soldier 3	(5,3)	(5,4)	1
Soldier 4	(7,3)	(7,4)	1
Soldier 5	(9,3)	(9,4)	1

Right cannon capture. $(8,2)\rightarrow(8,9)$, with the single screen at (8,7).

$b=3+2+2+2+2+2+1+1+1+13+10+5=44$

(B) One-ply leaf evaluations: 44

(C) Three-ply leaf evaluations: $b^3=44^3=85184$

If every non-root node is evaluated: $44+44^2+44^3=87164$

Game2 (p.2)

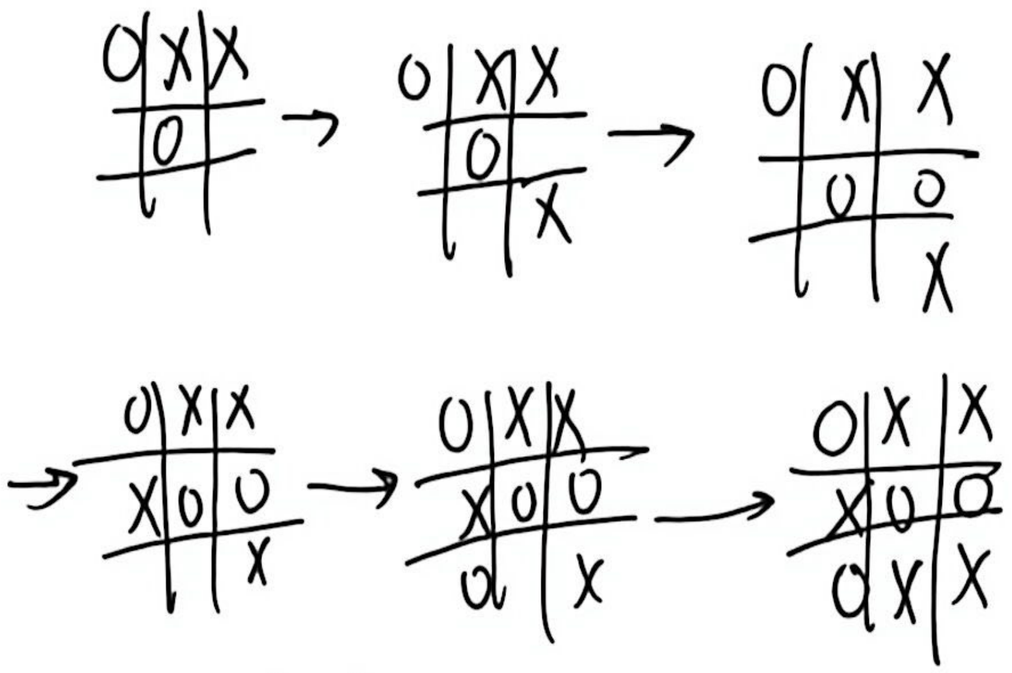
Cell numbering:

1 2 3
4 5 6
7 8 9

X first: alternating moves. Two legal continuations

First game. The supplied prefix is X2, O6, X3, O1. Add.

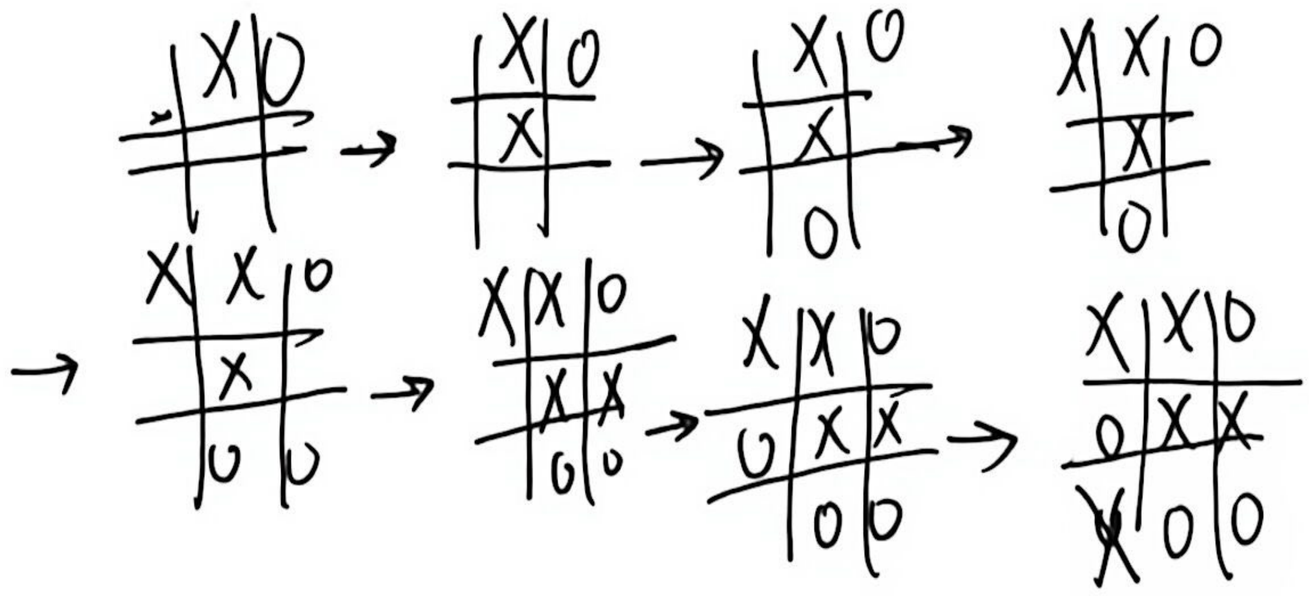
X9, O6, X4, O7, X8



Draw (no winning line).

Second game. The supplied prefix is X2, O3, Add.

X5, O8, X1, O9, X6, O4, X7



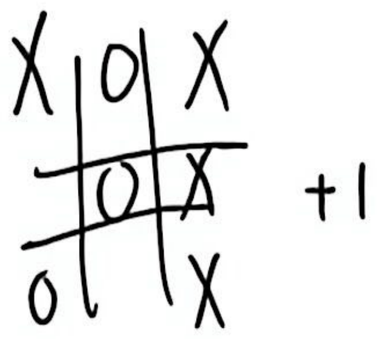
nowining line

Game3 (p.3)

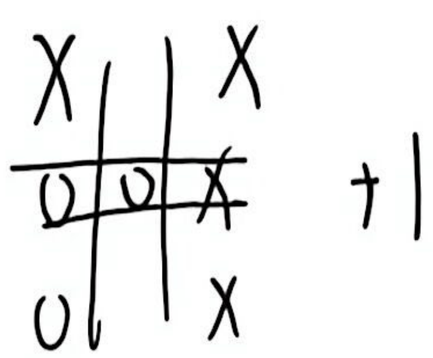
$V = +1$ (X win), -1 (O win), 0 (draw).

$$V(s) := \begin{cases} V(s') & s \text{ terminal}, \\ \max_{s' \in \text{children}(s)} V(s') & X \text{ to move}, \\ \min_{s' \in \text{children}(s)} V(s') & O \text{ to move}. \end{cases}$$

Terminal values:

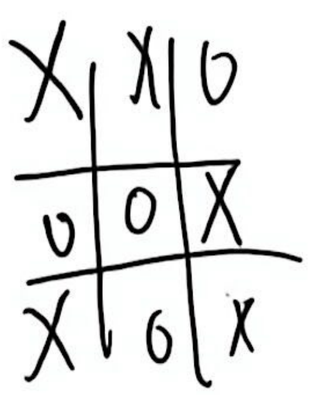


+1

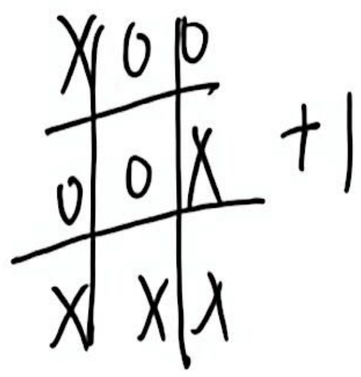


+1

Upper fork



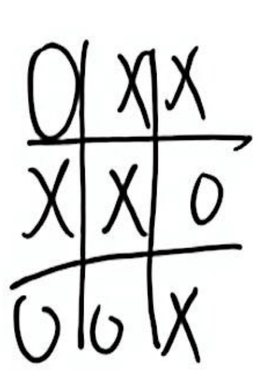
0



+1

Upper fork $\min(0,1)=0$; parent $\min(1,0)=0$. Single-successor chains retain the value.

Shared fork:



0



+1

Shared fork $\min(0,1)=0$

First X: corner. Pictured O replies:

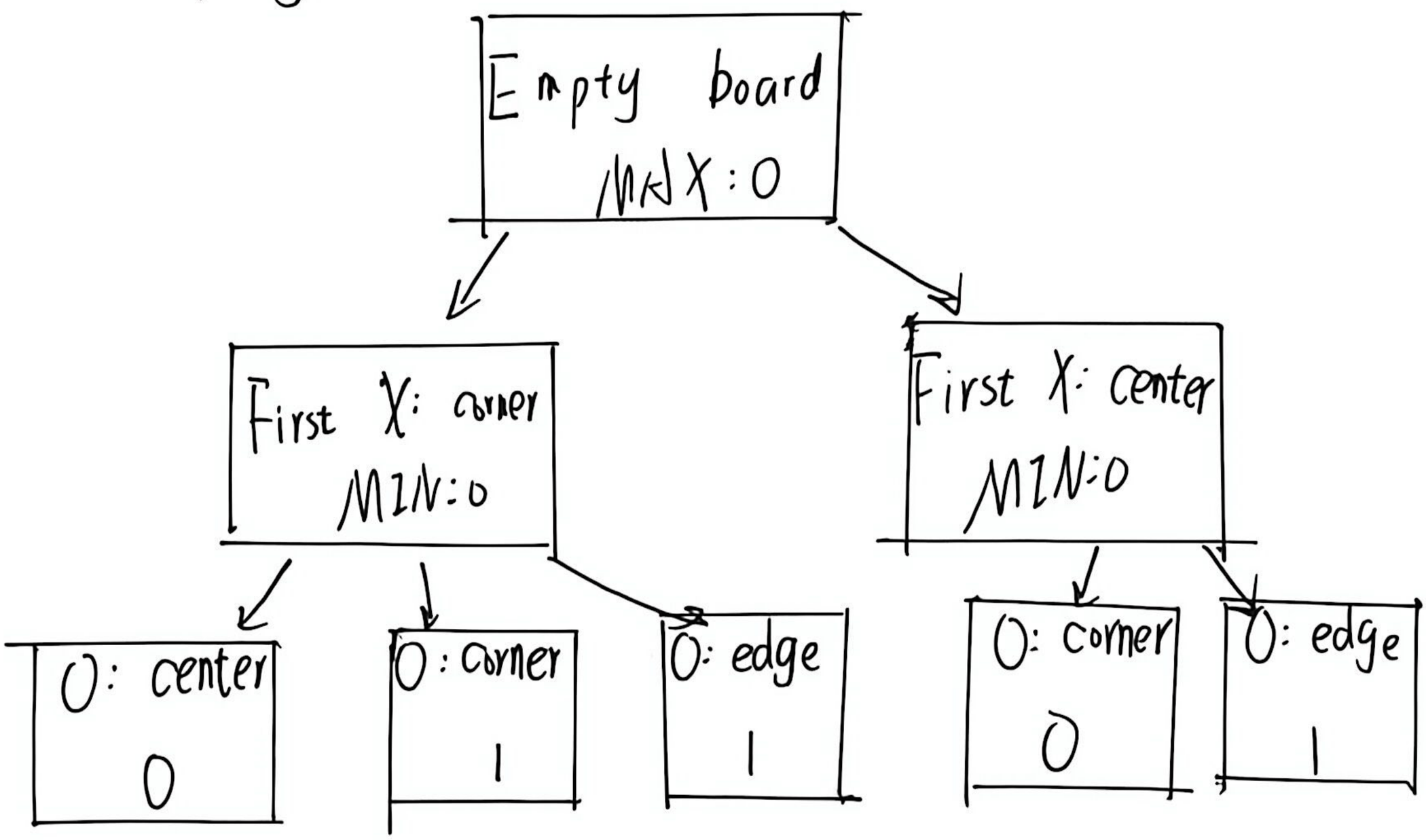
O in center: 0, O in top-right corner: 1, O on top edge: 1.

Dashed draw alternative at an X node: $\max(1,0)=1$.

Opening backup:

$V_{corner\ first} = \min(0,1,1) = 0$, $V_{center\ first} = \min(0,1) = 0$, $V_{root} = \max(0,0) = 0$

Opening tree



Choose center, Corner ties at value 0.

Values apply to the supplied partial graph. In the full game, perfect play draws from every opening.

Game 5 (pp.5-6)

X first; root depth 0; stop immediately after a win or a full board.

Leaves count move histories (no boord or symmetry merging)

Item	Reason	Result
1	Third X mark: move 5; full board: move 9	Leaf depths 5,6,7,8,9
2	Depth 5; X has 3 marks, O has 2.	$U_5 = +1$
3	Move 6 is O's ealier X wins already stopped	$U_6 = -1$
4	Move 9 is X's: win or full-board draw	$U_9 \in \{0, +1\}$

5. number of # depth-5 leaves. 8 wining lines; 3! orders for X;

C_6^2 ~~$\binom{6}{2}$~~ 2! placements/orders for O

$$N_5 = 8 \times 3! \times C_6^2 \times 2! = 1440$$

6. number of depth-6 leaves (optional). Count O wins, then exclude ealier X wins:

$$8 \times 3! \times C_6^3 \times 3! = 5760$$

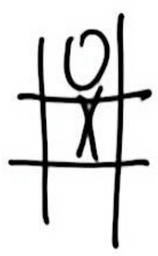
X. Disjoint ordered winning-line pairs: $3 \times 2 + 3 \times 2 = 12$.

Invalid orders per pair: $3! \times 3! = 36$

$$N_6 = 5760 - 12 \times 36 = 5328$$

Game 4 (p.4)

Initial board:



h(s) = # open X lines − # open O lines. MIN over O replies

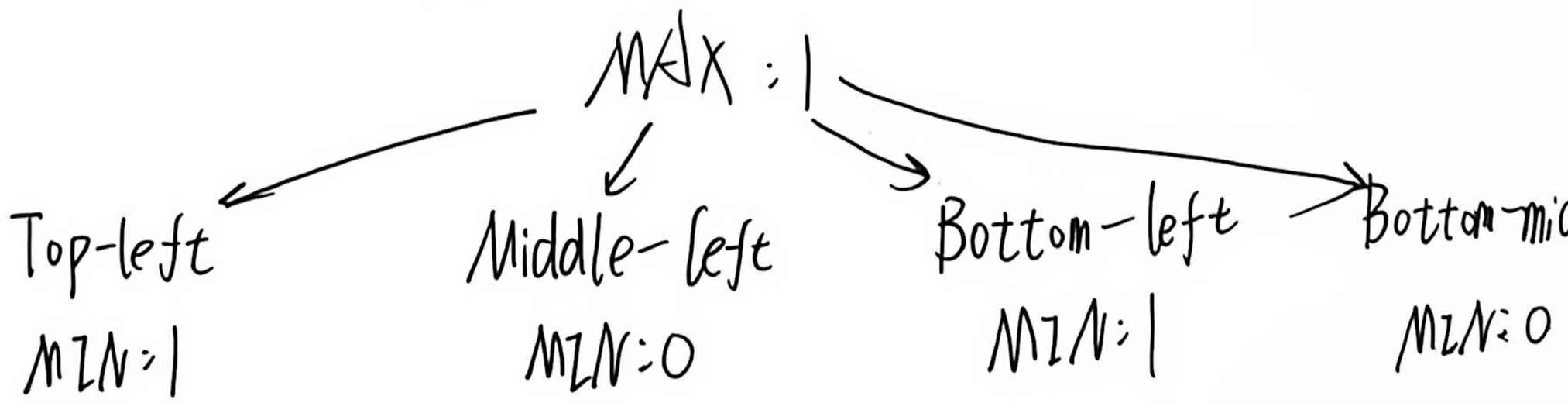
X's candidate move	Scores after O's responses	MIN value.
Top left corner	2, 1, 3, 1, 2, 1	1
Middle-left edge.	1, 0, 2, 0, 1, 1	0
Bottom left corner	2, 2, 3, 1, 2, 2	1
Bottom-middle edge	1, 1, 0	0

Root backup

max(1, 0, 1, 0) = 1

Best: all four corners

Choose bottom-left as a tie-break: same worst-case value 1, more favorable orther reply scores.



Chosen move:



7. Depth-q leaves (optional). $L_d(s,t)$ counts terminal continuations in exactly d further moves:

$$L_0(s,t) = \begin{cases} 1, & s \text{ is terminal}, \\ 0, & \text{otherwise}, \end{cases}$$

and for $d>0$,

$$L_d(s,t) = \begin{cases} 0, & s \text{ is terminal} \\ \sum_{a \in \text{legal}(s,t)} L_{d-1}(\text{Result}(s,a),\bar{t}), & \text{otherwise} \end{cases}$$

Enumeration of the unmerged legal tree:

Depth	X wins	O wins	Draws	Terminal histories
5	1440	0	0	1440
6	0	5328	0	5328
7	47952	0	0	47952
8	0	72576	0	72576
9	81792	0	46080	127872

Thus

$$N_9 = 81792 + 46080 = 127872.$$

Check: 16 full draw boards; every mark order is legal

$$16 \times 5! \times 4! = 46080.$$

Total terminal histories:

$$1440 + 5328 + 47952 + 72576 + 127872 = 255168.$$

$9!$ is invalid: it includes histories continuing after an ealier win.